

Music and code

Andrea Vigani

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Preface

This writing is not:

- intended to be a book or a manual.
- exhaustive.
- written in a descriptive/discursive style.
- aimed to provide absolute truths or certainties.

It aims to:

- provide readers with different aspects of electroacoustic music from different perspectives.
- prompt readers to think individually or collectively about specific topics.
- provide readers with input that will encourage them to delve deeper into the topics they find most interesting among those presented.
- help readers build their own poetic consciousness and learn the most suitable informatic tools to achieve it.
- help readers to find their own compositional models and procedures.

With the same purposes I use it as a framework in my University and Conservatory lessons.

It is always *under construction*.

Check first page to see the last date version.

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Chapter 1

Premises

1.1 Preliminary reflections

Compose something (music, furniture, poetry, novels, images, ideas, etc.).

What does it mean?

To compose, to assemble, to put together something.

Let's tackle some basic concepts about in no particular order.

1.1.1 Concrete things

If we want to produce a non-vocal sound we have to choose or assemble a physical object.



Then:

- we can interact with it (touch it, move it, etc.).
- we can make a copy (a model).
- we can modify it (improve it).
- we can destroy (or forgot) it.
- it does not require linguistic representation, it simply be.
- it exist in space and not necessary in time.

This has to do with the physical world.

Pyhsical object with elestic characteristics can be exited in some way producing sound waves.



Sound waves are a form of energy that propagates in time and space through an elastic medium.

Sound waves simply exist in our world.

When a sound wave reaches our ears it is perceived as an acoustic information by our brain.



It's a concrete thing.

We live immersed in sound waves.

We listen to music more or less daily but... is music always sound waves?

No.

We can think and organize sounds inside our mind without producing them as sound waves.



Sound waves have to do with physical world and human perception.

Music is a complex thing that has to do with both perceptual and cognitive human systems.

1.1.2 Pre-linguistic

We can compose ideas or instinctive sequences of sounds putting thoughts together in a pre-linguistic manner.



An abstract thing.

When we think it:

- we can't interact with it (touch it, move it, etc.).
- we can do a copy (replicate pattern structure).
- we can reproduce it but only in its physical forms (kinds of matter):
 - declaiming it (sound waves).
 - writing it on a paper using symbolic representations.
- we can modify it.
- we can't destroy it (we can only destroy its representation in physical world like book, recordings, etc., not in our memory).
- it does not require linguistic representation.
- it exists in the space of consciousness (our mind) but when we reproduce it in the physical world it exist both in space and time.

This has to do with mind, consciousness and human expression.

Let's explore this concept through a comparison between natural language and musical language.

They:

- are universal (present in all human cultures).
- belong exclusively to our species.
- ensure the cohesion of a social group.
- share formal and functional characteristics.

Natural language

Psycholinguistic studies say that at a deep level all natural languages have the same structure and this can tell us something universal about the human intellect (our brain).

The form of human thought is innate and common to all members of the species.

The function of language is to express it.



The deep structure of an expression is closely related to the thought it represents.

If pre-linguistic thoughts have the same type of form for everyone the linguistic structures that represent them must also have the same type of form.



If we want to transform a deep thought structure into a statement we have to assemble it together into a linear sequence of sounds.

This sequence must convey to the listener the necessary informations for its comprehension.

Thought itself cannot be considered a linguistic sequence: it exists independently of language.

Musical language

About music there exist forms of mental activity independent of any musical language such as [children's](#) babbling [1] or [tibetan](#) songs [2].

Pre-linguistic musical thought must be an abstract scheme that includes exclusively the characteristics common to all musics.

If both music language and natural language are universal characteristics of the human specie this means that humans have a natural capacity to acquire both linguistic and musical skills.

Natural language and musical language expressions are conveyed in the real world in the form of sound waves.

The sound waves common basic physical parameters are:

- frequency → interval, pitch, intonation.
- amplitude → dynamic.

- time → rhythm.
- timbre → instrumental techniques, vocal expressions, orchestration.

Even though surface forms differ from culture to culture we can find these universal elements not only in physical parameters but also in the subdivision into multiple levels of the languages.

1.1.3 Compose a phone number



Defining a code.

- a sequence of numbers (or letters or sounds) that represents a thought.
- we can't modify it.
- it requires encoding and decoding.
- to understand it we need to know its linguistic structure.
- it follow precise and shared rules.
- if we don't know them, the sequence of numbers or sounds makes no sense to us.
- it require linguistic representation.
- it can be:
 - in space → if you write it in a phone directory.
 - in time → if you digit it on a phone keyboard.

This has to do with language and its representations.

Let's start again with the comparison between natural and musical language.

Leaving aside the first common level represented by the physical parameters of sound both are built on further different levels.

Simplifying, we can identify four levels:

- phonetic-phonological → phonetics and prosody (notes, pitches, accents, tunings, etc.).
- morphosyntactic → combination of phonemes into morphemes and morphemes into words (rhythmic cells, chords, etc.).
- syntactic → rules defining the relationships between words (counterpoint, harmony, twelve-tone system, asides, phrases, etc.).
- semantic → access to meaning.

Phonetic-phonological level

It is present in all languages.

Every linguistic output can be broken down into phonemes (a small set of sound classes).

For example the word *ape* is made up of three phonemes one for each letter.

Phonemes have two distinctive features:

- they are not characterized by absolute values but by a set of sounds within a certain range.
- the continuum of sounds is divided differently in different languages (two sounds that constitute two phonemes in one language form a single phoneme in another).

THE INTERNATIONAL PHONETIC ALPHABET (revised to 2018)

CONSONANTS (PULMONIC) © 2018 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

In western musical languages the fundamental phoneme could be considered the note with its various types of expression (staccato, tenuto, accentato, legato, etc.).



In both languages they are characterised by frequency, duration, amplitude and timbre.

Morphosyntactic level

Phonemes can join together to form morphemes.

Morphemes can join together to form words.

Words are basic elements of language that:

- carries meaning.
- can be used on its own.
- are uninterruptible.

In music we could consider them as a short melodic pattern (inciso).



A vocabulary (lexicon) is a set of words in a given language.

In music we could consider it as a set of rhythmic and melodic patterns.

Substantial differences between the two languages:

- in natural language a word has a general function (the word *table* represent a table).
- in music a specific word exist only within a piece (we can consider Beethoven's fifth symphony incise a word only in this symphony).

Syntactic level

Syntax or formal grammar.

a closed system of rules that serves exclusively to generate the set of sequences considered grammatical.

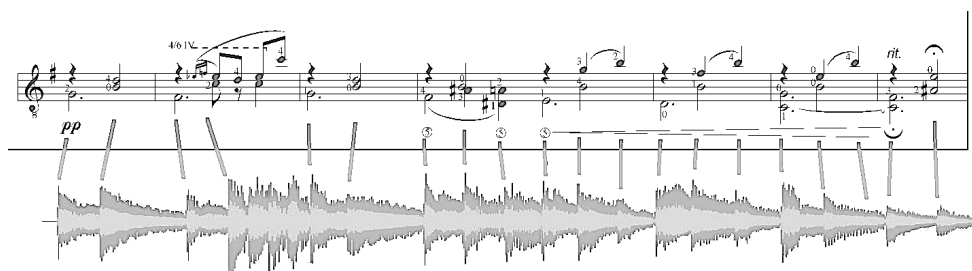
Every natural language like every musical language has its own syntax (italian, french, german, counterpoint, harmony, twelve-tone, etc.).

We have previously defined that music is not necessarily conveyed through sound waves but... are sound waves always music language?

No, again.

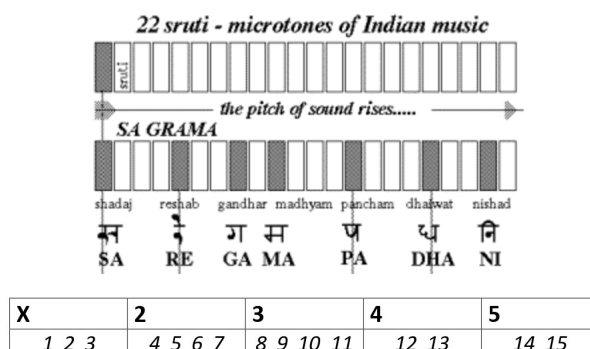
In order to be defined, music language requires:

- a codification of sounds in vocabularies.
- a systems of rules.



In all musical cultures the physical parameters of sound are organized in symbols.

These symbols can become independent from sound waves.



In Western musical tradition, from psamody until around 1940 these parameters have maintained more or less the same vocabulary over the centuries (diatonic and chromatic intervals, metric and subdivision of beat, musical instruments, etc.).

	PIU'CHE DIMINUITO	DIMINUITO	MINORE	GIUSTO	MAGGIORE	ECCEDENTE	PIU' CHE ECCEDENTE
PRIMA			MAI		MAI		
SECONDA				MAI			
TERZA				MAI			

The systems of rules through which the vocabulary terms were organized (syntax), however, were different (modality, counterpoint, harmonic systems, dodecaphony, alea, etc.).

- designative meaning → refers to something extramusical (G.Mahler - [X Symphony](#) [3]).
- embodied meaning → the meaning a piece has for the listener in terms of:
 - internal syntactic structure.
 - interactions between this structure and the listener's musical knowledge and expectations. Musical structure can create expectations that can be disappointed or fulfilled generating a dynamic flow of tensions and resolutions that influence the listener's emotional and aesthetic responses. These are aesthetic emotions that have nothing to do with the emotions experienced in real life (F.Schubert - [Trio](#) [4]).

Can we express in words what emotions listening to this Schubert trio arouses in us?

We will delve deeper into this last topic in the Music's meaning section ([1.2](#)).

These levels enable common phenomena that help us understand the difference between sound waves and language.

- categorical perception → a continuous linguistic or musical sound is segmented into discrete categorized units (phonemes, notes, words). If we repeat the same phrase or melody several times they will always be recognized as the same despite variations in acoustic parameters sometimes even significant ones ([example 1](#) [5], [example 2](#) [6], [example 3](#) [7]).
- phonemic restoration → replacing part of the linguistic signal with noise simultaneously with the word or music does not affect the perception of the whole. Semantic-lexical or musical expectations prevail over acoustic analysis ([musical](#) [7], [natural](#) [8]).
- expectations → the syntactic construction of a sentence leads us to expect one word rather than another within it, just as happens within a sequence of chords in tonal terms ([words](#) [9], [chords](#) [10]).

1.1.4 Perceive and process

The ability to perceive and process musical information is also common to all cultures.

Listening to music from one's own culture provides the listener with additional information.

The simplest musical competence requires the activation of numerous cognitive abilities:

- memory → short-term recognition of melodic rhythmic patterns.
- maintenance of attention → active listening.
- analysis of the temporal structure → continuous comparison between present and short-term memory.

These abilities are part of a body of knowledge acquired through more or less conscious implicit learning that occurs through two modalities:

- universal cognitive mechanisms → experimental psychology studies have shown that even prenatally the fetus responds to sounds and noises that activate phonological priming processes. For example, infants have been found to prefer sounds or stories frequently reproduced by their mothers after the third month of pregnancy over those they have never heard. Even in adulthood, some research has highlighted the existence of perceptual mechanisms independent of cultural context.
- environmental and cultural interaction → constant, quantitatively preponderant, continuous and more or less conscious perception of sounds organized according to the reference syntax of the cultural environment of reference.

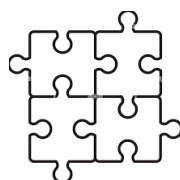


An example of the dichotomy just described can be found in analyzing the pitches of a melody.

We must activate several cognitive mechanisms which, in their simplest form are:

- processing the melodic profile (the progression of the ups and downs) → occurs more easily in both children and adults, with or without musical experience or literacy.
- processing musical intervals (the distance between pitches) → occurs less easily in general and easily in individuals with more experience or musical literacy ([melody \[11\]](#)).

1.1.5 Re-compose a puzzle



Re-constructing an image.

An abstract thing reconstructed in the physical world through a sequence of actions.

- we continuously interact between abstract and real world through attempts to correctly reproduce the original image.
- we can reproduce the process many times but we cannot consider them copies.
- we can't modify it.
- it does not require necessary linguistic representation but cognitive skills.
- during the re-composition process it need time and space but when the process is ended it need only space (in the case of a musical reproduction a space in our memory and consciousness).

This has to do with music representations, interpreters, players and performers.

Up to now we have thought about sound, music and their perception but not about the actors involved in the process of making music.

- composer.
- interpreter.
- listener.

We will delve deeper into these concepts in paragraph *Forms of human expression* ([1.3](#)).

1.2 Music's meanings

When we talked about the semantic level we mentioned the meaning but... what is the meaning of music and what does music represent for humans?

We start again with sound waves: they represent music in the physical world like clouds or mountains.

This has to do with both perceptual and cognitive processes.

By abstraction we can consider sound waves as [signs](#) [12].

Let's think on these topics:

- their production, transmission and interpretation.
- the ways in which something is communicated and signified.
- the production of symbolic objects.

1.2.1 Signs and sounds



Let's go hiking.

We're driving on a mountain road.

We see cars parked on the side of the road.

We realize that the mountain path we're looking for probably starts there.

We can consider the many parked cars a sign that indicated the beginning of the trail.

Our intentions, the context, and the tracks we encountered along the way activated within us an action-reaction mechanism made up of perceptions, expectations, emotions, interpretations.

We can define:

- the sight of the group of cars → the signifier (plane of expression) → what made us understand something.
- the presence of the beginning of the trail → the signified (plane of content) → what we understood.

The union of these two elements gives rise to a sign that generates an increase in knowledge.

Having found the beginning of the trail is what we can call the *paradigmatic effect* of the sign.

We were looking for something and we found it.

Listen ([example 1](#) [13], [example 2](#) [14], [example 3](#) [15]) and answer.

1. what kind of information do they provide us?
2. what is the signified content of these acoustic expressions?
3. what kind of increase in knowledge was achieved after listening to them?

These short sound texts could refer to three classic genres of electroacoustic music: soundscape composition, computer music and tape music.

All are carried by sound waves but have profoundly different signified contents that cannot be deduced (or composed) starting from the acoustic parameters of the sound.

1.2.2 Composers, players and listeners



Come back to the mountain path.

- we're walking on the path.
- the clouds are gathering → signifier.
- we understand that it's about to rain → signified.
- we decide to come back to the car → practical effect.

Signs often have a practical effect.

Sounds often have a practical effect (hearing thunder is like seeing clouds).

Have music practical effect?

Signs connect:

- something perceptual → the sight of cars or clouds with
- something cognitive → the presence of the path or impending rain.

Signs are not signifiers → their perception by someone is.

A sign is not something that represents something else.

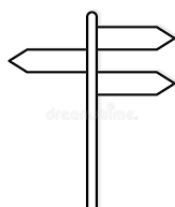
It represent the relationship someone establishes between two elements:

- a perceptible dimension (the view of the clouds).
- an intelligible dimension (we connect it to the possibility of rain).

This relationship:

- occurs *a posteriori*.
- is not necessarily intentional on the part of the sender (clouds are not harbingers of rain, just as the person who parked on the side of the road didn't mean to tell us the start of the trail).

We can connect signifier and signified only through an experiential factor (previously, when we saw those clouds, it rained).



Signification → as the examples just given, we are constantly surrounded by a multiplicity of unconsciously produced signs.

All these signs have the potential to become signifier expressions if interpreted by a recipient in the general act we can define as signification.

Communication → the sign is voluntarily produced by the emitter to convey a message, such as the signs in the previous image or the same images within this text.

Listen ([example 1 \[16\]](#), [example 2 \[17\]](#), [example 3 \[18\]](#)) and answer.

- what is the practical effect of the three sound texts?
- is sound information contained in the acoustic properties of the sound texts?

Sounds can be both signification and communication.

Music can be only communication.

In the performing arts of non-oral traditions (such as the western musical tradition), things get even more complicated because several actors play in the communication process:

- composer → usually thinks of a composition in his mind (symbolic musical language) and then represent it through some kind of notational code on a score.
- score → the coded instructions needed by an interpreter to reproduce the composer's ideas in the physical world in the form of sound waves.
- interpreter → must know both the musical language used by the composer and the symbolic language through which it was codified (semiography).
- listener → receiver that interprets an interpretation.

In acoustic music it is always like this, while in electroacoustic music things can change.

1.2.3 Do you know this music?



What is the relationship between signifiers and signifieds activated by the receiver based on?

Is this relationship always valid and somehow implicit in the signifier, or does it depend on context and conditions?

Let's consider the example of clouds and rain.

There is no an universal law that says that every time there are clouds in the sky, it will rain.

The relationship is established through *a posteriori* inductive generalization codified in an interpretative rule → it often rained after observing clouds in the sky, and therefore, whenever we encounter clouds, we associate them with the phenomenon.

Does this also apply to the example of cars parked on the side of the road?

In this case:

- there isn't an universal law according to which every time we observe parked cars, a mountain path is nearby.
- there isn't an interpretative rule established by inductive generalization, since parked cars rarely indicate the start of a trail.

The way of reasoning that led to the signification (inference) was induced:

- by the contingent dimension → place and context.

- by the affective dimension → enthusiasm for the hike, desire to find the trail to start walking, etc.
- by the reference culture, or that set of knowledge, value systems, habits, and behaviors within which we live and which have almost naturally led us to that conclusion and not another.

The cognitive inferences used daily in our interpretations are not entirely personal or subjective.

They are based on precise codes.

These codes:

- are formal systems that transcend individual choices.
- are not universal.
- are more or less lasting stabilizations of collective ways of thinking, acting, desiring.
- are dictated, maintained, and modified by social pressure.
- are social and cultural customs, interpretative habits that take on the appearance of a law (without actually being one).

If we hadn't shared the same culture and been part of the same society with the many people who parked their cars on the day of the excursion, that sign simply wouldn't exist.

The sign isn't in things or ideas, but in the forms of their relationship.

Listen ([example 1 \[19\]](#), [example 2 \[20\]](#), [example 3 \[21\]](#)) and answer.

Describe the differences between the three sound texts:

- what is the first information that comes to mind when you listen?
- what kind of information is?
- are there differences between the linguistic systems used in the three sound texts?
- if there are any, what information do they influence?

Some of these arguments lead us to understand sound signs as a cultural phenomenon and not merely acoustic since from this point of view the signifier coincides with the signified.

1.2.4 Do you like this music?



The above codes are anthropological conventions.

The relationships between expressions and content are connected to values, preferences, and tastes, all of which are extremely complex and are:

- arbitrary → when viewed from the outside.
- undisputed → when viewed from the inside.

A sign arises through difference when a sensorial discontinuity materializes, such as when, while driving along a mountain road, we encounter parked cars.

The perceptual gap brings to the surface something significant, important, which we value by distinguishing it from other signs encountered along the way.

We value it in two ways:

- we are happy because we have found what we were looking for, namely, the opportunity to embark on a beautiful hike in the mountains.
- we have achieved what we were seeking through a journey of knowledge that has kept our attention high (observing signs, the environment, other signs, etc.) and evaluating by comparing things (would it be better to stop at a restaurant or go for an hike?).

Social codes do not completely transcend individual choices but are linked to the ways in which the individual assumes them by mediating between collective and individual values.

In summary:

- value is what we aim for and what directs our series of actions and passions, giving precise meaning to each of them according to an implicit path.
- value arises from the evaluative comparison of things, objects, and from the recognition of differences; the greater the evaluative comparisons along the path, the greater the value.
- the continuous mediation between collective and individual values generates changes in anthropological codes.

Let's see a [video](#) [23].

1.2.5 Texts and narratives



The sign is just an iceberg tip.

It works because:

- it breaks down into many small, interrelated elements.
- it composes larger entities by relating to other similar signs.

The parked cars signify the discovery of the path because:

- their unusual number in that place.
- the way they are parked on the edge.
- they thicken as they approach the path and then thin out.
- they are in that place and not somewhere else.
- they take on value thanks to differential relationships with other signs.
- we are looking for the beginning of the path and not something else.
- all these and other elements are part of a story that unfolds in a precise period of time.
- this narrative is seen through a specific value-based, cultural, and anthropological perspective (a mountain hike).
- we are in a specific emotional state (positive tension generated by the anticipation of being able to begin something we enjoy).
- ...

The relationships established between these elements in a narrative generate the code through which that signifier refers to that probable meaning and not to another, forming what we might call a dynamic text.

Signs function because they are intertwined in texts.

The words of a language are like signs:

- the variable result of constant relationships between smaller elements (morphemes, phonemes, sound features)
- themselves entities that compose larger structures (sentences, texts, speeches).

Every larger element capable of meaning transcends the smaller elements.

The meaning of a sentence transcends the meaning of the individual words that compose it, as well as the individual phonemes.

Texts are not just books, documents, etc., but any portion of the world with:

- determined limits (context).
- precise internal articulation (code).

that carries some configuration of meaning, or better yet, a signification.

Do you remember the puzzle?

In order for meaning to:

- be produced.
- be circulated.
- be received.
- be transformed.

it must refer to texts or units of meaning that various societies, historical periods, and cultures use to define their core values.

Let's see a [video](#) [22].

1.3 Forms of human expression

We can summarize the concepts just presented by following the thoughts of the musicologist J.J.Nattiez set out in his text *Music and discourse*.

Forms of human expression (including music) can be defined as symbolic forms only if we recognize three levels:

- poietic dimension (emitter) → the set of strategies activated by the author that lead to the creation of the work (something that did not exist before).
- aesthetic dimension (receiver) → the set of strategies activated by the listener during the perception of the work.
- the work itself (neutral) → the text with its own internal organization that can be analyzed independently from the poietic and aesthetic dimensions.



1.3.1 Poietic dimension

Active process of construction by the emitter of interpretant signs (composer).

Each sign does not refer directly to an object but makes use of the intermediary action of a second sign which is its interpretant in a process of multiple references.



For example, in verbal language, if we write a word like *table*, we have:

- created a graphic sign (grapheme → formed by a sequence of letters and defined by its own rules) that, if
- correctly interpreted by the reader, will cause him or her to pronounce the corresponding word (morpheme → a sequence of phonemes defined by its own rules). If
- correctly interpreted by the hearer, it will refer to a general idea of a table (an abstract concept → defined by conventions derived from collective experience). If
- correctly interpreted by someone, it will connect it to a real, meaningful table (object → defined by subjective experience).

This is true for a single word.

The interpretative signs become infinite when individual words are organized into a language.

Ognuno sta solo sul cuor della terra trafitto da un raggio di sole: ed è subito sera

Let's try to imagine the same process in music and the amount of control the composer can have over the [sound information](#) [24] he wants to transmit.

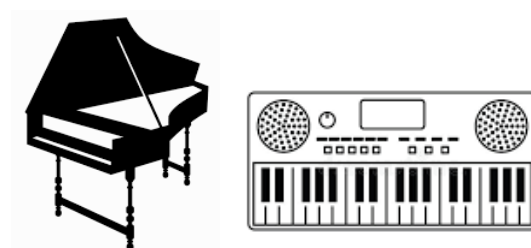
1.3.2 Aesthetic dimension

Active process of construction by the interpretant.

Process of understanding that can be defined as sharing information on cultural operations typical of a human social group.

In the case of artistic languages, the interpretative signs attributed to the work by the emitter are not necessarily the same as those projected by the receiver because:

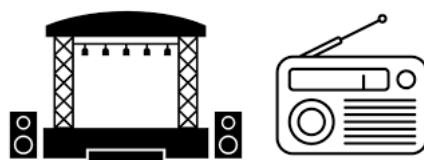
- they are modulated by the context in which the interpreter perceives the work.
- they are modulated by any discrepancy in period or cultural context between the emitter and the receiver.
- the perception of a sonata in the Baroque era is presumed to be totally different from the perception of the same work today, as the historical, social, technological, and cultural contexts are significantly different.



The perception of a Beethoven symphony by an Amazonian native is totally different from ours, as is our perception of the microtones present in the [muezzin chants](#) [25].



Let's also think about the technological dimension: the perception of the same sound text in a live concert is different from the perception of its reproduction recorded by an electroacoustic system perhaps while we wait for the subway.



1.3.3 The work itself

Regarding the understanding of the sound text itself Nattiez advocates the need to analyze:

- works and styles as cultural and anthropological objects.
- harmony, rhythm, meter and all aspects that have a similar function in the grammar of natural language (comparison).

When we speak of grammars of musical parameters we are not referring exclusively to the analysis of the formal and structural characteristics of a specific musical text. We must also take into consideration the grammars of musical parameters generalized and historicized in styles and forms.



Nattiez borrows from the study of verbal language the semantic exercise of breaking down meaning according to the use of words, which in music corresponds to the analysis of a melodic profile or a rhythmic system, or the distribution of harmonic fields in a tonal system or twelve-tone series, etc.

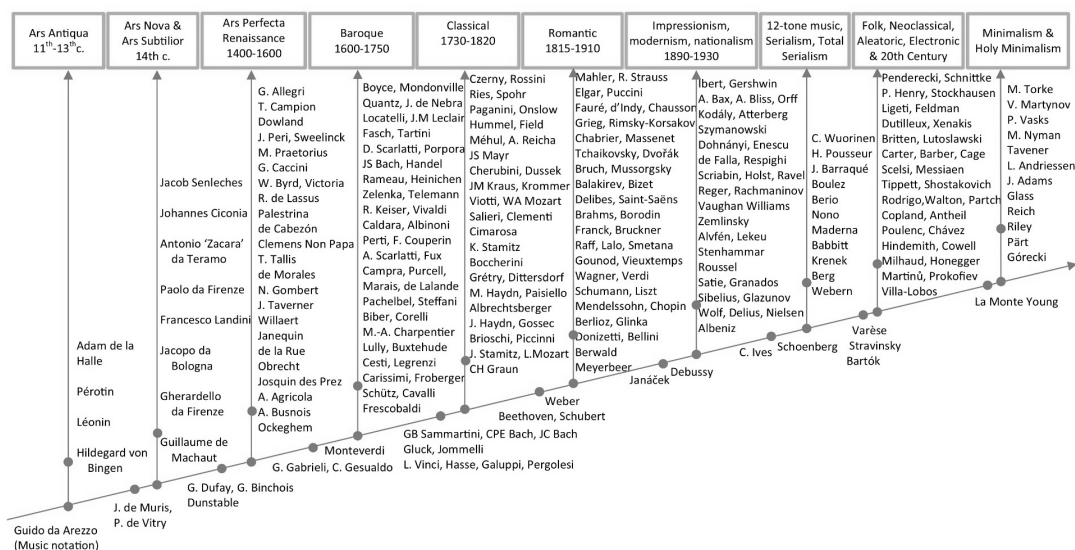


Each musical element (pitch, intensity, duration, timbre) can be separated and taken as a strategic variable in the production of a [musical work](#) [26].

For Nattiez constructing a musical theory is the organization of a selection of interpretants adopted within an anthropological context.

Anyone who wants to invent a new musical grammar (a practice that has been widespread among art music composers since the 1950s) must take this into account and not just the definition of rules internal to the text.

Works do not exist in isolation since it is their place in history that provides their meaning and possible interpretations over time.



The sound text can live a life of its own and be isolated from the the emitter but not from the anthropological context in which it was created and/or reproduced.

1.3.4 Final reflections

Everyday life is filled with more or less voluntary signification.

Signification is the amniotic fluid within which human beings live.

We can highlight it and clarify its mechanisms.

This task is often hidden in everyday life.

If we ask someone why they like a piece of music or a dress they will probably answer:

- because it's beautiful.
- because it excites me.
- because it's right.
- because. . .

Meaning often hides behind other motivations that tend to suppress it.

Let's think about written and spoken language.

In using it we put into practice extremely complex knowledge that we tend to be unable to explain.

We would hardly be able to fully explain all the grammatical, phonetic, morphological, and syntactic rules we apply daily.

Language is practical knowledge that we almost never reason about.

Our daily experience works the same way: it appears random, spontaneous, and immediate.

In reality it is based on a dense and constantly changing network of relationships between things, people, places, etc.

Also the act of composing a piece of music should develop in the continuous passage from awareness of linguistic mechanisms (intelligence) to creativity.

Let's see a [video](#) [27].

Chapter 2

Genres, instruments and tools

2.1 Genres

Before starting to write an electroacoustic piece we should:

- define the instrumentation to be used which can include acoustic instruments, electroacoustic instruments, virtual instruments or a mix of these types.
- define the sound amplification and diffusion system.

For this reason we categorize the different types and genres of electroacoustic music by dividing them into two macro areas defined by the way of controlling sounds and musical parameters.

- non interactive systems.
- interactive systems.

We will explore the compositional strategies of each in dedicated chapters.

2.2 Non interactive systems

This category includes genres that:

- don't involve any human interaction during the performance.
- involve interactions that don't influence the musical or sound text in any way.

According with the reflections of the previous chapter we can affirm:

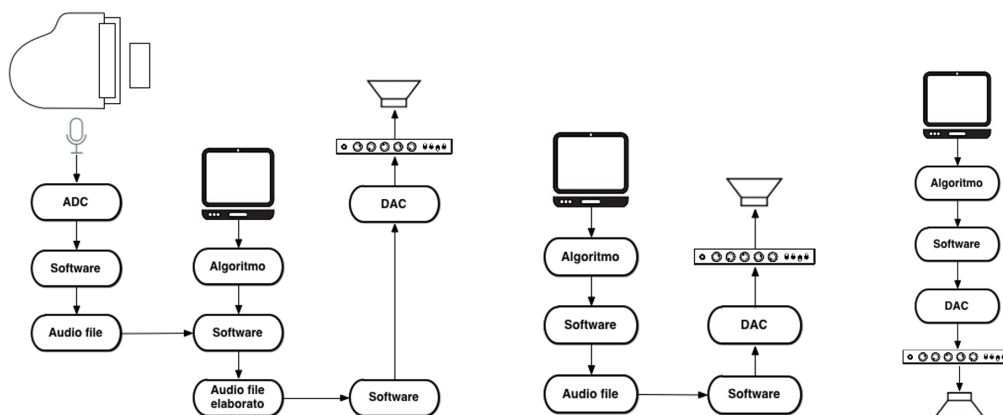
- interactions may influence the listener's perception of the sound text (mainly through the different characteristics of the diffusion systems used).
- interactions don't change the sound text.

This category can be divided into two further subcategories:

- fixed media music ([2.2.1](#)).
- live sequencing ([2.2.2](#)).

The image shows the possibles audio chains in this category.

- non-real-time sound processing.
- non-real-time synthesis.
- real-time synthesis.



2.2.1 Fixed media music

This subcategory denotes music or video that is played back from a recording.

This used to be called *tape music* but tapes are no longer used.

The entire work is stored on a single medium and reproduced from beginning to end.

The scores of the pieces belonging to this category are secondary to the sound text and obtained through various sound analysis techniques.

They are defined as listening scores (and not performance scores) and are represented in the form of sonograms or oscillograms.

Musically, over the years, various musical analysis techniques have been developed for this type of sound texts.

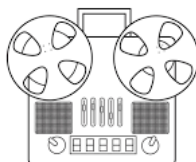
As examples of theories of analysis we can download:

- a [text](#) by Laura Zattra.
- a [bulletin](#) of the gatm edited by Marco Ligabue.

In this subcategory we can identify three main genres:

- acousmatic music.
- wallpaper music, ambient music and soundscapes.
- radiodrama.

Acousmatic music



We can start from *musique concrète*.

In the 1940s Pierre Schaeffer began composing with the sound (acoustic world).

After applying various sound elaborations (editing, filtering, reversal, etc.) he began to apply musical techniques to their combinations (repetition, transposition, reordering, etc.) inventing new musical structures (symbolic world).

In doing so, he reversed the compositional practice that had been in force for centuries in the western musical tradition.

The listeners could not make the cognitive associations illustrated in the first chapter.

They activating exclusively the basic cognitive mechanisms common to all human beings.

This listening mode has been called *reduced listening*.

All listeners are comparable to the aborigine who listens to a Beethoven symphony.

P.Shaeffer - [Etude aux chemins de fer](#) for tape (1948) [28] - [Analysis](#).

- mono analog tape.
- recorded and processed train sounds.
- sounds from the acoustic environment.
- we can recognize the sound source and its processing.
- perceived as a distortion of reality.

Acousmatic music is the modern evolution of *musique concrète*.

By the mid-1970s a number of composers felt a need to designate their conception clearly in terms of specific methodology, syntax, and tools.

François Bayle suggested adopting the term *acousmatique*.

This term comes from the greek *akousma*: what is heard.

Students of Pythagoras listened to his teachings from behind a screen unable to see him.

He believed that the lack of visual cues would force his students to focus all their attention on his message.

After his death his followers split into:

- acousmatics (practitioners of the mystic doctrine)
- mathematics (scientists).

The loudspeaker was the modern equivalent of the screen partition.

"From an esthetical point of view acousmatic music concentrates on the poetical and spectral richness of sounds, and plays with this very particular characteristic of sound hearing in which the perception of anacoustic phenomena is associated with its cause; hence the perception of a sound whose cause is unknown or unrecognizable for our perception, induces the listener to imagine non-existing causes and to perceive music as a complex creative phenomena in which musical sense and musical sounds have to be interpreted simultaneously, with generally very little relation with our perceptive reality. The question is not to find out how sounds are made but how their combination will generate imaginary perceptions of imaginary realities in our mind".

Teruggi, D. 1995. What about acousmatics? In *Journal of Electroacoustic Music*. Vol. 7. London: Sonic Arts Network - [More info](#).

B.Parmegiani - [De Natura Sonorum](#) for tape (1975) [29] - [Analysis](#).

- stereo digital tape.
- recorded and processed sounds plus synthetic sounds.
- sounds from an imaginary environment.
- we cannot recognize the sound source.
- perceived as an alternative or virtual reality.

Another characteristic of acousmatic music is that it is performed live on loudspeaker orchestras called *acousmonium*.

Indeed Bayle referred to acousmatic music as:

"art of projected sounds which is shot and developed in the studio, projected in a hall, like cinema".

H.Vaggione - [Consort](#) for convolved violins (2011) [30] - [Analysis](#).

- multitrack digital file.
- recorded and processed instrumental sounds.
- sounds from music environment.
- we can recognize the sound source and its avatars.
- perceived as a piece of music.

[More info](#) about acousmatic works interpretation.

Wallpaper music, ambient music and soundscapes



We can start from Eric Satie (non-electroacoustic composer).

He was a precursor of minimalism, muzak, and many other 20th-century musical genres.

Satie conceived this musical genre as a backdrop to everyday activities (passive listening).

Satie was familiar with this idea due to its role as a Montmartre café pianist in the late 19th and early 20th century.

E.Satie - [Musique d'ameublement](#) (1917) for ensemble [31].

- acoustic instruments.
- pattern iteration.
- music environment.
- time dilation (psychological time).
- passive and non immersive listening.
- perceived as a piece of music.

We know well how nowadays we are bombarded daily by this type of music in a totally involuntary motion.

Let us recall what was stated in the first chapter on the basic cognitive mechanisms.

In the 1950s a visionary composer like J.Cage took this idea to its extreme with the piece 4'33'' whose sonic and musical content consists of the background noise of a finite time.

Later in 1978 another composer Brian Eno coined the term *ambient* as a musical genre.

A Sunday morning in the Cologne airport while waiting for a flight he thought:

The light was beautiful, everything was beautiful, except they were playing awful music. They spend hundreds of millions of pounds... on everything. Except the music.

It was this that compelled him to begin composing music for public environments.

We must consider that the diffusion of sounds and/or music in environments not designed for active listening can change the perception of that environment in a subconscious way.

It is time that through sound redraws the boundaries of a space.

B.Eno - [Music for Airports](#) (1978) for tape [32].

- stereo vinyl.
- pattern iteration.
- music environment.
- time dilation (psychological time).
- passive and semi immersive listening.
- perceived as a piece of music.

[More info](#)

Soundscape composition can represent a meeting point between musique concrete and ambient music because:

- it uses sound materials recorded from the real world or synthesized.
- does not recompose them according to symbolic rules and principles into a musical form.
- its purpose is to construct or reconstruct a real or virtual soundscape.

It's essentially based on the studies and works of canadian musicologist and composer Raymond Murray Schafer.

To put it simply, we can say that different types of sounds coexist in a sound environment and we can identify three types:

- geophony → all the sounds of nature of non-biological origin.
- biophony → all the sounds of nature of biological origin but not caused by humans.
- anthrophony → all sounds generated by humans.
 - language.
 - music.
 - mechanical sound.

Schafer also divides the sounds of a soundscape into two categories:

- hi-fi → all sounds that have an hight signal-to-noise ratio (they are distinguishable from the background noise).
- lo-fi → all sounds that have a low signal-to-noise ratio (they are indistinguishable from the background noise).

Three main elements in a soundscape:

- keynote → sounds that most characterize an environment, analogous to the tonality in tonal music.
- sound signals → all the foreground sounds that do not necessarily characterize an environment but may appear occasionally.
- soundmark → The unique sounds of a given soundscape.

N.Barret - [Soundwalk](#) (1998) [33].

- stereo digital tape.
- no pattern.
- acoustic environment.
- time as in real life (clock time).
- passive and immersive listening.
- perceived as an acoustoc environment.

[More info](#)

Radiodrama



Radio drama can use only four kinds of signs:

- speech.
- sound effects.
- music.
- silence.

Any one of these by itself can be a very slippery client to deal with.

Silence can lead a listener to suspect a transmission failure and switch off or tune over to another one.

Sound effects are notoriously misleading - a sneeze can be taken for a bomb explosion!

An unfamiliar music will sound irritatingly like cacophony.

And speech says nothing to someone who doesn't know what you mean by the words.

But if you hit on a harmonious combination of the right choices of these four kinds you can move mountains, deploy infantry battalions with Air Force support, immerse a soul in the joys of paradise.

In short do anything you please, all in few minutes one after another.

The main focus is narrativity and a continuous balance in the construction of meaning between explicit sounds (both in natural and auditory language) and perceptual hints and/or allusions.

O.Wells - [War of the worlds](#) (1938) [34].

- mono radio signal.
- natural language.
- didactic sound design to simulate the real acoustic environment.
- time as in real life (clock time).
- active listening non immersive.
- perceived as a representation of reality.

[More info](#)

Computer music - algorithmic composition



The birth of computer music is naturally linked to the invention of the electronic calculator and the subsequent spread of personal computers as well as the invention of new (formal) languages that allow humans to communicate with these machines.

Starting from the second half of the 1950s, two orientations emerged:

- the first one (developed mainly in Europe) deals with the digital coding of musical parameters (assisted composition, assisted musical analysis, etc.).
- the second one (developed mainly in USA) deals with the digital coding of sound parameters (synthesis and processing).

In 1957 MUSIC I was created by Max Mathews in Bell Laboratories.

It was the first software for playing music from mathematical functions.

After MUSIC I, there were: MUSIC II, III, IV, V, MUSIC 360, and then C-sound and Super-Collider languages still used today.

The main sound synthesis and processing techniques used are:

- additive synthesis.
- subtractive synthesis (filters).
- FM (Frequency Modulation) synthesis.
- granular synthesis.
- analysis and resynthesis (FFT).
- generative algorithms for:
 - musical structures.
 - timbres.
 - pitches.
- remote control of computers (MIDI, OSC, HID).

As we all know, nowadays computer is the main musical instrument, like piano in the 19th century.

The term *computer music* as an aesthetic category identifies the musical genre adopted by the pioneers from the mid-50s to the late 90s

J.Chowning - [Stria](#). (1977) [35].

- stereo digital tape.
- sonic environment.
- sounds from imaginary environment (only synthetic sounds - FM).
- time dilation (psychological time).
- active listening.
- perceived as a piece of computer music.

[More info](#)

2.2.2 Live sequencing

This subcategory is almost identical to the previous one except for two aspects:

- works can be generated in real time through algorithms (computer music).
- tracks are divided into different sound files triggered in real time (sound design for theatre and gaming).

Sound design for performing arts or video



Works on fixed media which differ from the works illustrated in the previous section for:

- sound does not have an absolute value but must be functional to dramaturgy.
- divided into more or less short fragments.
- open form → fragments can be triggered in non-linear sequences (slightly different than what was said about random generation).

Anonymous - [Dance!](#) (2010) [36].

- stereo digital tape.
- music functional to movement or dramatic timing.
- secondary sensitive plane.
- passive listening.
- perceived as a complement to visual information.

Sound design for gaming



In this case too the sound must be functional to the game

It should:

- suggest a mood, evoke a feeling.
- indicate a geographical locale.
- define a character.
- mirror or exaggerate how things sound in real life.
- clarify the narrative

To simplify there are five sounds categories:

- dialogue → any verbal speech in the game (player talk).
- music → any non-diegetic music (orchestral music).
- sound Effects (Hard Effects) → any sound from an real-life object (sounds of rocks falling).
- foley → any sound effect that the player makes (footsteps, etc.).
- backgrounds (ambience) → noise from the environment (wind, rain, etc.).

Procedural audio:

- soundfiles triggered by user actions.
- real time shynthesis by internal oscillators.

Artistically we could merge game music, soundscape composition, acousmatic music and radio-drama.

Various - [8 bit music](#) (1990) [37].

- stereo digital samples or real time oscillators.
- music functional to gamer emotions.
- also called procedural music.
- pattern iteration.
- music environment.
- time dilation (psychological time).

- non-immersive audio diffusion but immersive psychological function.
- passive listening.

[More info](#)

2.3 Interactive systems

This category includes genres that:

- involve human interaction during the performance as happens with acoustic instruments.
- these interactions influence the musical or sound text at different levels.

Compared to the previous typology the presence of one or more performers influences:

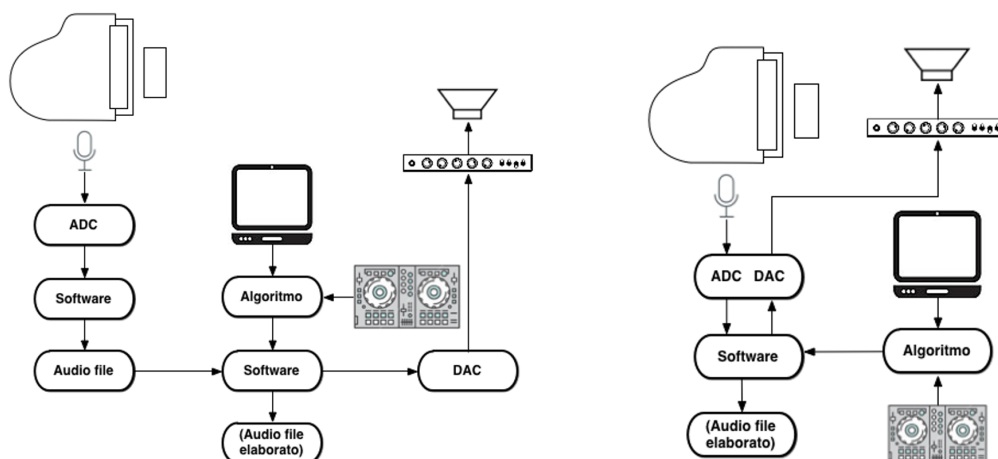
- the anthropological and contextualised perception of the musical action.
- its representative nature.

This category can be divided into three further subcategories:

- live set.
- hyper-instruments.
- live coding.

The image shows the possibles audio chains in this category.

- real-time processing of pre-recorded or pre-synthesized sounds (parameters can be controlled through various types of devices - MIDI, OSC, HID, sensors).
- real-time processing of signals captured through microphones or sensors (parameters can be controlled in the previous modes but also through control signals derived in some way from the incoming audio signal).



2.3.1 Live set



Slork - [Twilight](#) (2013) [38].

Live performance using analog or digital electronic musical instruments.

Often musicians improvise in a live set, making each performance unique.

Can be a solo performance or a laptop ensemble.

One of the most important aspects of this genre is to think about and assemble original performance environments which leads to the search for new human/machine interfaces dedicated to the generation of new sounds in the musical field.

We should design integrated organisms understood as networks of humans, musical instruments, and machines where the former communicate with the latter through *creative embodied interfaces* (sensors, microphones, cameras, smartphones, etc.) in the continuous search for new real and virtual luthiers (and related gestural techniques of instrumental performance) oriented to the changing expressive needs of our time.

We must find the right balance between what humans are good at doing and what computers are good at doing, trying to overcome their respective limitations in an attempt to reach and explore new artistic goals.

For every project we must ask ourselves:

- does the aesthetic-artistic aim of the project justify the technology used?
- is there at least one expressive element in the project that can't be represented exclusively through traditional musical instruments?
- is human-machine interaction intrinsic to the generative idea of the project, so much so as to involve the design of specific executive techniques linked to both real and virtual instrumental gestures?

The same questions apply to projects similar to the next paragraph.

2.3.2 Hyper-instrument



P.Boulez - [Anthèmes 2](#) (1994) [39].

Augmented musical instrument that uses technology to extend the capabilities of a traditional instrument and enhance the performer's expressiveness.

Two types:

- real-time augmented instrumental techniques → sound processes are controlled by:
 - an electronic performer who interacts with different types of controllers.
 - by information extracted from the audio signal produced by the instrument itself (feature extraction as control signals).
- cyborg luthiery → sound processing is controlled by movements and/or gestures of the performer captured by different types of sensors.

Three main goals:

- respond to the performer's input in a dynamic and expressive way allowing for a deeper connection between the musician and the music.
- produce sounds and musical effects that are impossible to achieve with traditional instruments alone.
- expanding musical forms.

2.3.3 Live coding



SuperCollider [Live Coding](#) [40].

Also called:

- on-the-fly programming.
- just in time programming.
- conversational programming.

Search for new musical forms, new places for performance and new audiences.

It is part of a larger artistic movement called *creative coding*.

If channeled into the western musical tradition, we can think of it as a new version of organ improvisations in the baroque era.

From a semiographic point of view the most interesting aspect is that the score is the code and is created and manipulated in real time.

The compositional process which has always belonged to the author's private sphere becomes public and an integral part of the performance itself (do you remember the first paragraphs of the previous chapter?).

2.4 The virtual instrument paradigm

In this section we define a virtual instrument model that will be useful in next chapters.

If anyone is unfamiliar with SuperCollider, they can download an introductory text at [this link](#) or read the *Appendix ??*.

In the chapter on live electronics we'll make some small modifications to allow signal input.

2.4.1 Software installation

Download and install [SuperCollider](#).

We can copy and paste the code into the SuperCollider IDE Interpreter from this file or download and launch the code from link at the beginning of each section.

2.4.2 Instrumental model

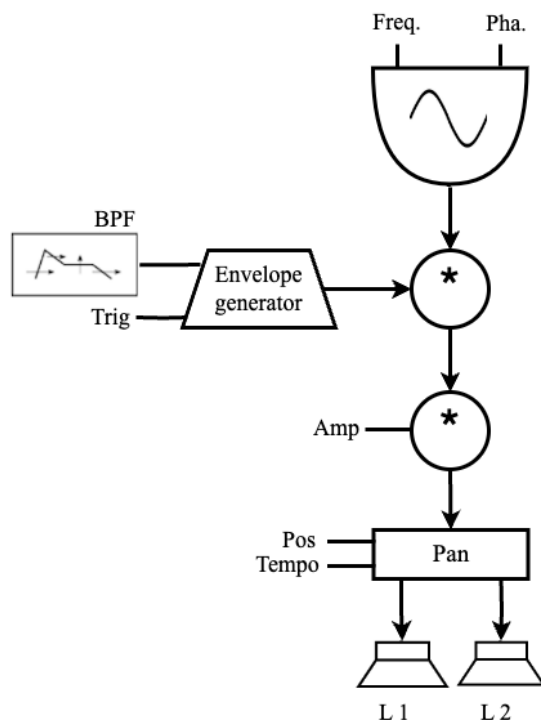
[Download](#) chapter's SuperCollider code.

Let's create a computer model that contains functions common to most virtual instruments (sound generators).

- a source signal (oscillator or mic/line input).
- an amplitude envelope (with or without sustain).
- a stereo panner.
- a signal output.

We also establish the basic parameters we want to control dynamically (they will increase depending on the synthesis algorithm chosen).

- frequency (Hz).
- general amplitude (0.0 to 1.0).
- duration (seconds).
- pan position (-1.0 to +1.0).
- pan interpolation time (movements)
- output bus.
- trigger.
- kind of voice allocation (doneAction:0 or 2).



We find a more detailed description of the different components, connections and parameters of this virtual instrument model in Appendix [??].

```

1 s.boot;
2 s.meter;
3 s.scope;
4 s.plotTree;

```

In SuperCollider we can define it with the *SynthDef* (synthesis definition) class.

This class contains the instructions (algorithms) needed to build multiple copies (instances) derived from the model.

```

1 SynthDef(\model, {arg freq=500, amp=0, dur=1, pan=0, pantime=0.0,
2   out=0, t_gate=0, done=2;
3   var sig, env;
4   sig = SinOsc.ar(freq);
5   env = Env.new([0.0,1.0,0.5,0.7,0],
6     [0.1,0.1,0.3,1.0].normalizeSum * dur,
7     -4
8   );
9   env = EnvGen.kr(env, t_gate, doneAction:done);
10  sig = sig * env * amp;
11  sig = Pan2.ar(sig, pan.varlag(pantime));

```

```

12         Out.ar(out, sig)
13     }).add;

```

Listing 2.1: Instrument model

Now we can generate as many instances as we want.

We have two ways to control the instance parameters.

They vary depending on the voice allocation we want.

Dynamic voice allocation (poliphonic)

Instance self-kill when the amplitude envelope expires (*doneAction:2*).

In this case we pass parameters at instantiation time (array of | *key, value*).

```

1 Synth(\model, [\freq, rrand(890,1234),
2               \amp, 0.5,
3               \dur, rrand(0.1,2),
4               \pan, rand2(1.0),
5               \pantime, 0.1,
6               \done, 2,
7               \t_gate,1 ])

```

Listing 2.2: Dynamic voice allocation

Static voice allocation (monophnic)

Two passages:

- create instance and assign it to a variable.
- send one or multiple parameter changes by *set* method.

```

1 a = Synth(\model, [\done, 0]);
2
3 a.set(\freq, rrand(890,1234),
4       \amp, 0.5,
5       \dur, rrand(0.1,2),
6       \pan, rand2(1.0),
7       \pantime, 0.1,
8       \t_gate,1)

```

Listing 2.3: Static voice allocation

Then we can kill it or kill all Synths (Nodes) on the Server.

```

1 a.free;
2 s.freeAll;

```

The type of voice allocation is strictly linked to the type of amplitude envelope [??].

2.5 Composition sketches proposal

Create at least one of the following instrumental models by searching in the UGen browser (Help files) for anything not known.

- – source → white noise signal.
- – envelope → without sustain, percussive.
- – voice allocation → dynamic poliphony.
- – source → *Blip* UGen.

- number of harmonics → controlled by a LFO.
 - envelope → trapezoidal with sustain.
 - voice allocation → static.
 - pan → controlled by smoothing values from Interpreter.
- poliphony → two homorhythmic independent voices.
 - source voice 1 → Pulse wave.
 - source voice 2 → Sawtooth wave.
 - envelope → both adsr.
 - pan voice 1 → stereo left.
 - pan voice 2 → stereo right.
 - voice allocation → static.

Appendices

Listings

2.1	Instrument model	38
2.2	Dynamic voice allocation	39
2.3	Static voice allocation	39

Bibliography

- [1] <https://www.youtube.com/shorts/jq7CD9XM-ow>
- [2] <https://youtu.be/8-wLLhyVlLk?si=Pw-01Znj1LhWVgpK>
- [3] <https://www.youtube.com/watch?v=QVoviGYThxw>
- [4] <https://www.youtube.com/watch?v=e52IMaE-3As>
- [5] https://www.youtube.com/watch?v=QpEit_L1l0k
- [6] https://www.youtube.com/watch?v=_LCy21ddnsM
- [7] <https://youtu.be/NZUCq5GfsQ4?si=Vd2MxGYYxkB2p80z>
- [8] <https://www.youtube.com/watch?v=lwFGVv9h80o>
- [9] https://www.youtube.com/watch?v=qENwEuD__2E
- [10] https://www.youtube.com/watch?v=a_B02BZp-5Y&t=923s
- [11] <https://youtu.be/wmTHiCq8T3I?si=tC57HEpMAcVUa1MD>
- [12] <https://youtu.be/PZpH9Kh0E0?si=B8aluq7HSpKjwHe&t=77>
- [13] <https://www.youtube.com/watch?v=29XymHesxa0>
- [14] <https://youtu.be/iWoI8vmE8bI?si=JXVcAo10k-94U0BF>
- [15] <https://www.youtube.com/watch?v=-uIvzVgk16c&t=3s>
- [16] <https://www.youtube.com/watch?v=x0U605Wqen4>
- [17] https://www.youtube.com/watch?v=K2h88z_u_UQ
- [18] https://youtu.be/JQGCecX2d_Q?si=9YsT7CghkmdF-6bU
- [19] <https://youtu.be/TLDflhhdPCg?si=MWjDfFMNpnAjUnBz>
- [20] https://youtu.be/KQiBIb_klT8?si=AN7eCU0i_bQvdPii
- [21] <https://youtu.be/UEWCCSuHsuQ?si=KJuMp6i48CeySpBD>
- [22] <https://www.youtube.com/watch?v=Ie0wanLy5MI&t=3s>
- [23] <https://youtu.be/i1X3mz-k0oE?si=KL1os8wzWwQCjrIR>
- [24] <https://youtu.be/MHFuaaGpGKQ?si=F0LcXKWqlv3wL4h0>
- [25] <https://www.youtube.com/watch?v=G96FEkkFCzg>
- [26] <https://www.youtube.com/watch?v=7e0aIiHB58U>
- [27] <https://www.youtube.com/watch?v=asOFi1W228M>

- [28] <https://youtu.be/N9p0q8u6-bA?si=4ALUHn8MCJ0kuM1l>
- [29] https://www.youtube.com/watch?v=c_JHjUFf0s8&t=12s
- [30] <https://www.youtube.com/watch?v=F20kvxuXuEO>
- [31] <https://www.youtube.com/watch?v=Qwv0xJ-4Saw>
- [32] <https://www.youtube.com/watch?v=vNwYt1lyt3Q>
- [33] <https://www.youtube.com/watch?v=xf1JjUtnHs8>
- [34] <https://www.youtube.com/watch?v=Xs0K4ApWl4g>
- [35] <https://www.youtube.com/watch?v=988jPjs1gao>
- [36] <https://www.youtube.com/watch?v=VstAnd5GA6w&t=2276s>
- [37] <https://youtu.be/oMgQJEcVToY?si=glmYv4q5Pb0oTWe7>
- [38] <https://youtu.be/chA-4GRcb-I?si=VZUQ835EY9Tutgxk>
- [39] <https://www.youtube.com/watch?v=TMdDgwNLY8>
- [40] https://www.youtube.com/watch?v=Ix2b_qFYfAA&t=163s
- [41] <https://youtu.be/CIPTgie3Wjc?si=T6FxI8Lj6ynsKx7I>
- [42] <https://www.youtube.com/watch?v=8ftsFsScFbM>